

Amritaraj Nair

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EDUCATION

Texas A&M University — B.S. Computer Science Honors, Minor in Mathematics

Expected May 2029

Relevant Coursework: Data Structures & Algorithms, C++ Design, Discrete Structures, Engineering Calculus III, Python Programming

EXPERIENCE

Matic | *Software Engineer Intern*

Jun 2026 – Aug 2026

- Saved physicians **90+ minutes per provider daily (3 more patients seen per day)** by designing and shipping **Pulsematic**, an AI-powered inbox triage and auto-response drafting system for **athenaOne** and other **EHR inbox platforms**.
- Improved response relevance and reduced manual edits by architecting a **multi-agent orchestrator on Google Cloud Platform (GCP) with Vertex AI** to route, prioritize, and optimize automated drafting across heterogeneous message types.
- Reduced release turnaround time across services by building automated **CI/CD pipelines with Google Cloud Build triggers**, streamlining deployment velocity.
- Secured protected health information at scale by engineering a **PHI-compliant data management pipeline on Firestore**, enforcing strict access controls and data-handling safeguards.

MagNet Agents | *Software Engineer Intern*

May 2025 – Aug 2025

- Enabled automated client-acquisition workflows by engineering **data pipelines and web scraping systems** to extract and structure high-value legal lead data from public sources.
- Improved engagement and conversion by developing **AI-powered outreach generation features** using LLMs to create personalized messaging and legal opportunity summaries.
- Supported a Cornell-backed platform generating **\$50K+ ARR** by building core **backend systems for lead matching and tracking** that connect lawyer profiles with relevant prospects.
- Shipped **end-to-end product features** in a fast-paced startup environment at the intersection of legal tech and AI.

AI Modeling for Bio-Economic Systems | *Undergraduate Research*

Aug 2025 – Present

- Optimized livestock nutrition under economic and environmental constraints by applying **agent-based modeling (Mesa)** and machine learning.
- Improved reproducibility, scalability, and cross-pipeline integration by migrating legacy **R workflows to Python**.
- Evaluated sustainability and production outcomes by building cross-platform simulation systems in **Python, Java, and NetLogo**.

Maroon Fund — Scholars of Finance | *Quantitative Developer*

Feb 2025 – Present

- Developed **quantitative models and screening frameworks** for equity research across a **\$70K+ student-managed fund**.
- Improved investment-analysis efficiency by engineering **data pipelines and valuation tools** on financial datasets.
- Delivered data-driven insights for stock-pitch processes and portfolio decisions by automating **research workflows**.

PROJECTS

ClinicalHours | *CTO*

Dec 2025 – Present

- Placed in the **top 10% of the Spring 2026 Y Combinator batch** for building an **AI-powered email and call automation system (Gmail API, GoHighLevel)** that acts as a virtual receptionist, scheduling clinic meetings and accelerating application workflows for pre-med students.
- Enabled geolocation-based clinic discovery and automated communication pipelines by integrating **MapBox, Google APIs, and Resend API**.
- Transformed large-scale **U.S. hospital datasets** into structured, production-ready formats in Python.
- Shipped premium AI resume and application tools using **OpenAI and Gemini**; partnered with clinics including **BCS Free Health Clinic**.

Shot Sensei | *Co-Founder & Lead Developer*

Mar 2026

- Won the **Startup Ready Award at Hook'em Hacks 2026 (UT Austin)** for a computer-vision platform for competitive and training-based pickleball.
- Built a real-time mode where users play against an AI bot, plus a training mode using **Gemini** and **ElevenLabs** to analyze and coach all strokes (serve, volley, forehand, backhand).
- Implemented pose and shot detection with **OpenCV** and **YOLOv8** for stroke classification and performance feedback, backed by **Supabase** for win/loss records and shot-level scoring analytics.
- Invited to the **McCombs School of Business (UT Austin)** to pitch Shot Sensei to **Pear VC (Khalil Fuller)** as both a playable AI-powered game and a personalized training system.

AlphaForge | *Founder & Solo Developer*

March 2026

- Built **AlphaForge**, a full-stack quantitative investing platform to design, test, and deploy trading strategies, featuring a multi-level **Quant IDE** with a Scratch-like builder for beginners and an advanced multi-language coding environment.
- Improved strategy validation and iteration by implementing **backtesting and paper-trading systems**, plus a real-time market tracker for stocks, ETFs, indices, and crypto with integrated analytics and news.
- Integrated **Yahoo Finance, Alpaca, and Seeking Alpha APIs** and deployed on **Vercel** for scalable, low-latency performance.

TECHNICAL

SKILLS

&

HONORS

Languages: Python, C++, Java, TypeScript, SQL

Frameworks: React, Flask, FastAPI, Pandas, NumPy, Matplotlib

Tools: Git, Docker, AWS EC2/IAM, Google Cloud, Supabase, CI/CD

Awards: YC Internship Expo invitee · UT Austin Hook'em Hacks 2026 **Winner** (Most Startup Ready + Multimodal Track) · TidalTAMU Google Gemini Track **1st Place** (2026) · Outstanding Undergraduate Researcher Award · President's Endowed Scholar · Google Labs Makeathon · FRC 4192 Impact Award